

Yihao (Q) Quan

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 [Personal Page](#) |  [Github](#) |  [Google scholar](#) |  [Linkedin](#)

Education

- **Rutgers University - New Brunswick** Sept 2025 - Present
Ph.D. Student in Computer Science
- **Beijing Jiaotong University** Sept 2021 - July 2025
B.Sc. in Management Information Systems

Research Interests

My primary research lies in **MLLMs/LLMs**. Specifically:

- **Vision + Language:** Bridging the gap between language and vision representations, with the goal of building more interpretable and controllable multimodal systems.
- **Hallucination, Factuality, and Safety:** Leveraging interpretability insights to improve downstream capabilities such as factuality, reasoning, hallucination mitigation, and model editing, while also designing safer models.
- **Agentic AI:** Developing agentic AI systems with reliable memory, adaptive planning, and joint learning between agents and world models.

Selected Publications

** Equal Contribution*

- **[COLM 2026] Reinforcing Consistency in Video MLLMs with Structured Rewards**
MLLMs Video Understanding
Yihao Quan, Zeru Shi, Jinman Zhao, Ruixiang Tang.
[arXiv](#)
- **[EMNLP 2025, Oral, 325/8174 $\approx$ 3.9%] Shallow Focus, Deep Fixes: Enhancing Shallow Layers Vision Attention Sinks to Alleviate Hallucination in LVLMs**
MLLMs Hallucination
Yihao Quan*, Xiaofeng Zhang*, Chaochen Gu, Chen Shen, Xiaosong Yuan, Hao Cheng, Kaijie Wu, Shaotian Yan, Jieping Ye.
[Code](#) [arXiv](#)
- **[NAACL 2025, Oral, 226/3332 $\approx$ 6.7%] From Redundancy to Relevance: Information Flow in LVLMs Across Reasoning Tasks**
MLLMs Visual Reasoning Interpretability Efficiency
Xiaofeng Zhang*, Yihao Quan*, Chen Shen, Chaochen Gu, Xiaosong Yuan, Shaotian Yan, Liang Xie, Wenxiao Wang, Hao Tang, Jieping Ye.
[Code](#) [arXiv](#)
- **[ACL 2026 Main, Oral Recommended] ART: Attention Replacement Technique to Improve Factuality in LLMs**
LLMs Factuality
Ziqin Luo*, Yihao Quan*, Xiaofeng Zhang, Xiaosong Yuan, Chen Shen.
[Code](#) [arXiv](#)
- **[ECCV 2026] Counting Circuits: Mechanistic Interpretability of Visual Reasoning in Large Vision-Language Models**
MLLMs Visual Reasoning Interpretability
Liwei Che*, Zhiyu Xue*, Yihao Quan*, Benlin Liu, Zeru Shi, Ruixiang Tang, Ranjay Krishna, Vladimir Pavlovic.
[arXiv](#)
- **[NeurIPS 2026] AgentLens: Interpretable Safety Steering via Mechanistic Subspaces for Multi-Turn Coding Agent**
Agentic Safety Interpretability
Weidi Luo, Qiming Zhang, Yihao Quan, Mingyu Jin, Jie Cai, Chaowei Xiao, Jingcheng Niu, Zhen Xiang.
[Code](#) [arXiv](#)

- **[NeurIPS @ ResponsibleFM]** Read the Scene, Not the Script: Outcome-Aware Safety for LLMs
LLMs Safety Interpretability
 Rui Wu, Yihao Quan, Zeru Shi, Zhenting Wang, Yanshu Li, Ruixiang Tang.
Code arXiv
- **[AAAI 2025]** Enhancing Multimodal Large Language Models Complex Reason via Similarity Computation
MLLMs Visual Reasoning Efficiency
 Xiaofeng Zhang*, Fanshuo Zeng*, Yihao Quan, Zheng Hui, Jiawei Yao.
Code arXiv

Selected Preprints

* Equal Contribution

- **[In Submission]** Preserve Content, Disrupt Structure: Patch-Shuffle Counterfactual Distillation for Visual Reasoning
MLLMs On Policy Distillation
Yihao Quan, Jiang Liu, Ximeng Sun, Jialian Wu, Ruixiang Tang, Emad Barsoum, Zicheng Liu.
arXiv
- **[In Submission]** Why MLLMs Struggle to Count: Overcoming Individuation and Aggregation Bottlenecks with ConvStack
MLLMs Visual Reasoning Thinking with Image
 Liwei Che*, Yihao Quan*, Sen Fang, Hongyi Wang, Ranjay Krishna, Ruixiang Tang, Vladimir Pavlovic.
arXiv
- **[In Submission]** Improving Visual Reasoning with Iterative Evidence Refinement
MLLMs Visual Reasoning Thinking with Image
 Zeru Shi*, Kai Mei*, Yihao Quan, Dimitris N. Metaxas, Ruixiang Tang.
arXiv
- **[In Submission]** MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory
Agentic Safety Interpretability
 Minghao Guo*, Qingyue Jiao*, Zeru Shi*, Yihao Quan, Boxuan Zhang, Danrui Li, Liwei Che, Wujiang Xu, Shilong Liu, Zirui Liu, Mubbasir Kapadia, Vladimir Pavlovic, Jiang Liu, Mengdi Wang, Yiyu Shi, Dimitris N Metaxas, Ruixiang Tang.
Code arXiv
- **[In Submission]** The Anatomy of a Fake Citation: Field-Specific Hallucination Neurons in Large Language Models
LLMs Hallucination Interpretability
 Yuefei Chen, Yihao Quan, Xiaodong Lin, Ruixiang Tang.
arXiv

Selected Projects

- **Awesome-LMMs-Mechanistic-Interpretability (*Projects Lead*)** 200+ stars
A curated collection of resources focused on the Mechanistic Interpretability (MI) of Large Multimodal Models.
- **Awesome-World-Models (*Projects Contributor*)** 3000+ stars
A curated collection of resources focused on World Models (WMs).

Industry Experience

* Publications Listed Include Only (Co-)First Author Works

- **AMD GenAI Team** Jun 2026 – Present
 Research Topic: Multimodal On Policy Distillation
 Advisor: Jiang Liu
 Preprint: *In Submission*
- **Alibaba** Jun 2024 – Sep 2024
 Research Topic: Vision Hallucination Mitigation
 Mentor: Chen Shen & Xiaofeng Zhang
 Publication: EMNLP'25 (Oral)

Research Experience

* Publications Listed Include Only (Co-)First Author Works

- Rutgers University** Sep 2025 – Present
Research Topic: Video Understanding Enhancement, Mechanistic Interpretability
Advisor: [Ruixiang Tang](#)
Publication: In Submission, [COLM'26](#), [ECCV'26](#)
- KAUST (King Abdullah University of Science and Technology)** Sep 2024 – Nov 2024
Research Topic: Inefficient Attention Allocation in LVLMs
Mentor: [Yang Shu](#)
- Shanghai Jiao Tong University** Feb 2024 – Jun 2024
Research Topic: Information Flow in LVLMs
Mentor: [Xiaofeng Zhang](#)
Publication: [NAACL'25 \(Oral\)](#), [ACL'26 \(Main\)](#)

Professional Service

Reviewer: ICLR'27, COLM'26, ICLR'25, AAAI'26, AAAI'27, ARR Oct, ARR July, ARR May

Selected Awards And Honors

- COLM'26 Travel Award** 2026
- SIAM'26 Travel Award** 2026